

Problem 1 (Spring 2025). Let $n \geq 3$ and let $i \in \{1, 2, \dots, n\}$.

- (a) Show that $H = \{\sigma \in S_n : \sigma(i) = i\}$ is a subgroup of S_n .
- (b) In S_8 , let $\alpha = (1\ 2\ 3)(4\ 5)(6\ 7)$. Determine the number of elements in $C_{S_8}(\alpha)$, the centralizer of α in S_8 .

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